Masiingäänttär

Neural Machine Translation Between Standard Finnish and Rauman Giäl

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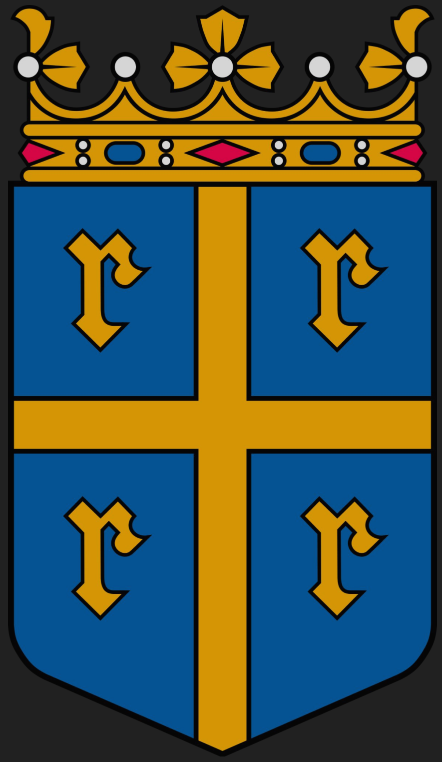
Project Description

- *Rauman Giäl* (Rauma) is a low-resource dialect of Finnish with unique terminology and morphology
- No corpus of Rauma data exists
- I synthesized a bilingual corpus of Finnish and Rauma data using web scraping, dictionaries, and GPT-3.5
- I trained a Finnish to Rauma neural translation model by finetuning an OPUS model on my Rauma data
- I expect finetuning to provide decent results that both use Rauma morphological endings and terminology



Rauma Coat of Arms

Related Work

- "Look It Up: Bilingual Dictionaries Improve Neural Machine Translation"
  - Attaching dictionary definitions of rare words before feeding through the encoder
- "Point, Disambiguate and Copy: Incorporating Bilingual Dictionaries for Neural Machine Translation"
  - Novel three-part architecture leverages bilingual dictionaries for disambiguation tasks
- "Dict-NMT: Bilingual Dictionary based NMT for Extremely Low Resource Languages"
  - Replaces words with a bilingual dictionary, and employs a neural model to smooth the translation
- "Domain-Specific Text Generation for Machine Translation"
  - Backtranslate source text from in-domain target text to improve domain-specific translation
- "Machine Translation for Low-resource Finno-Ugric Languages"
  - Finetuning pre-existing models is a prevalent technique
- "Revisiting Low-Resource Neural Machine Translation: A Case Study"
  - In the ultra-low data condition, reducing the BPE vocabulary size is very effective

Data Sources

Segments	
Bilingual Data	
Nortamo Society website	78
Rauma wiki	245
Monolingual Data	
Letters from Rauma blog	5,286
Helsinki Rauma Society proceedings	1,192
The works of Hjalmar Nortamo (3 books)	6,496
Raumalais Knekku Toti ja Jekku by Onni Helkiö	2,056
Total Segments	15,353

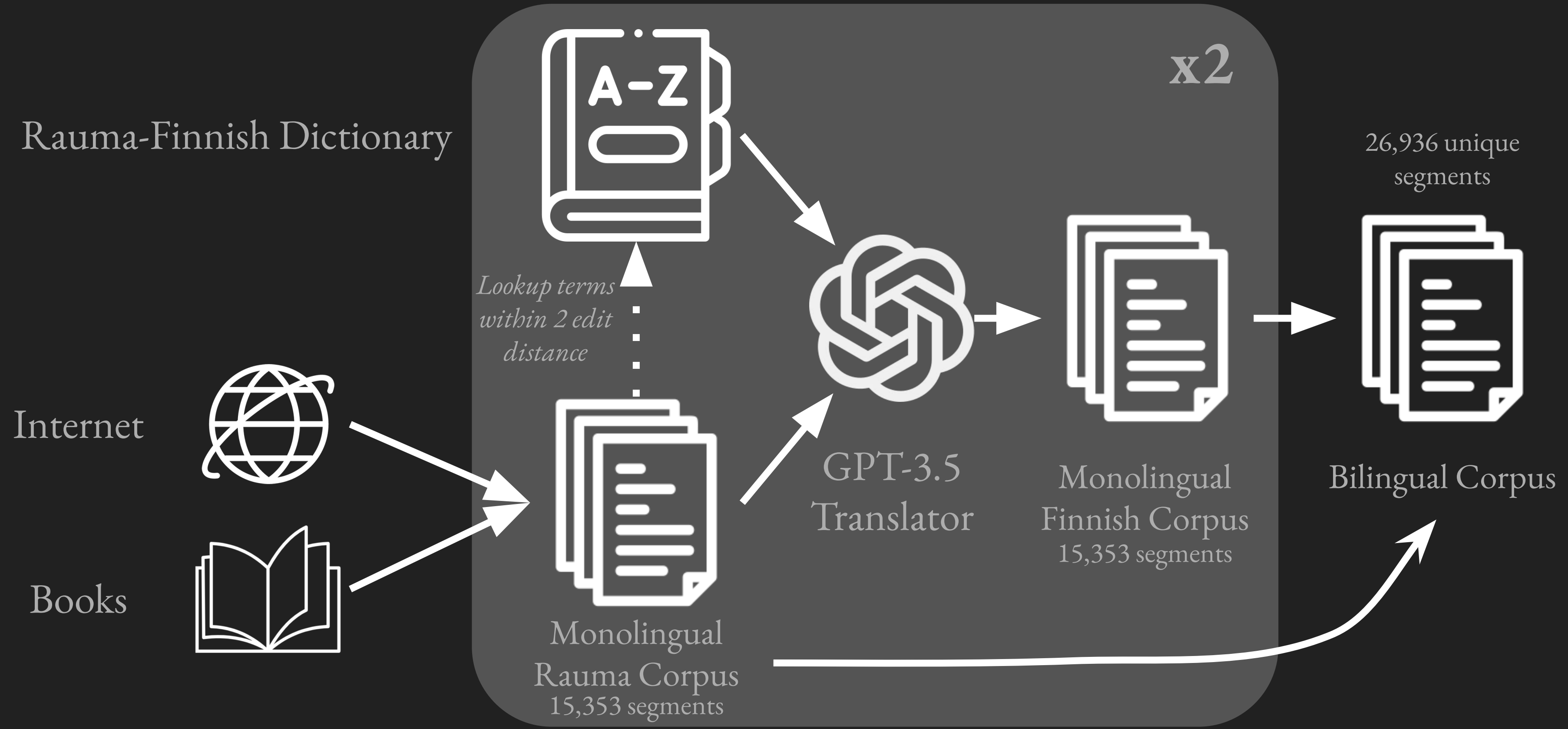
Rauma-Finnish Dictionary (4067 entries)

Frans Hjalmar Nortamo
Rauma Author

Code and Tools

- Web Scraper browser extension
- Data cleaning pipeline for web scraped data and dictionary
- Synthetic data generation with GPT API and dictionary
- Subword tokenization with SentencePiece
- Finetuned OPUS MT Finnish to Finnish model

Synthetic Data Generation Method



Synthetic Data Generation Example

Key:
Rauma
Finnish

I want you to translate from Rauman Giäl dialect to standard Finnish. Here are some definitions that might help:

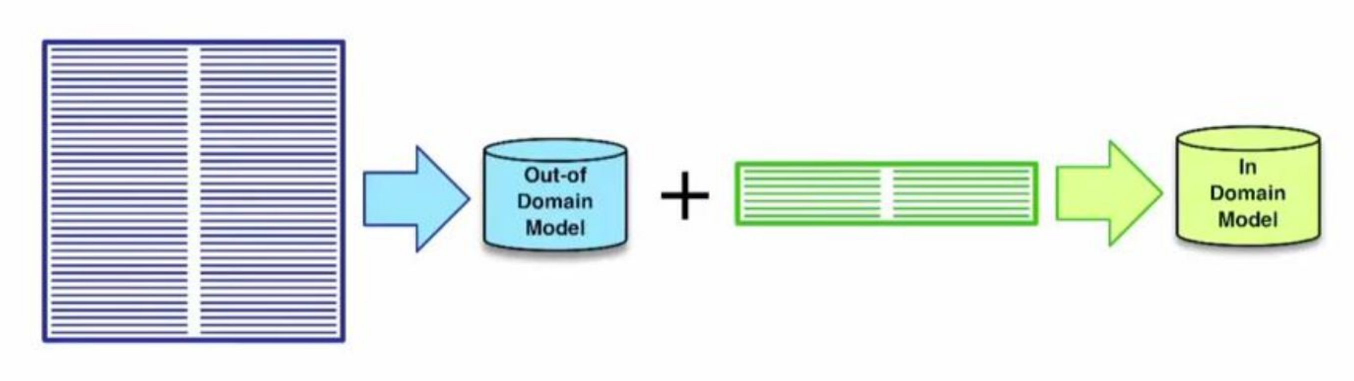
<u>aaholli</u> veto- ja jarrutusmenetelmä, jossa köysi kierretään puun ympärille
traction and braking system, where a rope is wrapped around a tree

Rauma: Täyty pan <u>aaholli</u>, etei pääs luiskattama.

Finnish (sans dict): Täytyy panna <u>varovasti</u>, ettei pääse liukastumaan.
Have to put it <u>carefully</u> so that it doesn't slip.

Finnish (with dict): Täytyy panna <u>ankkuri</u>, ettei pääse liukastumaan.
Have to put <u>an anchor</u> so that it doesn't slip.

Model Finetuning with OPUS MT



OPUS Dataset (Finnish to Finnish)	opus-mt-fi-fi MarianMT Model	Synthetic Dataset (Finnish to Rauma)	Masiingäänttär MarianMT Model

Training: 26,536 segments
Validation: 400 segments
Test: 323 segments


Demo
Freistaus

Results and Evaluation

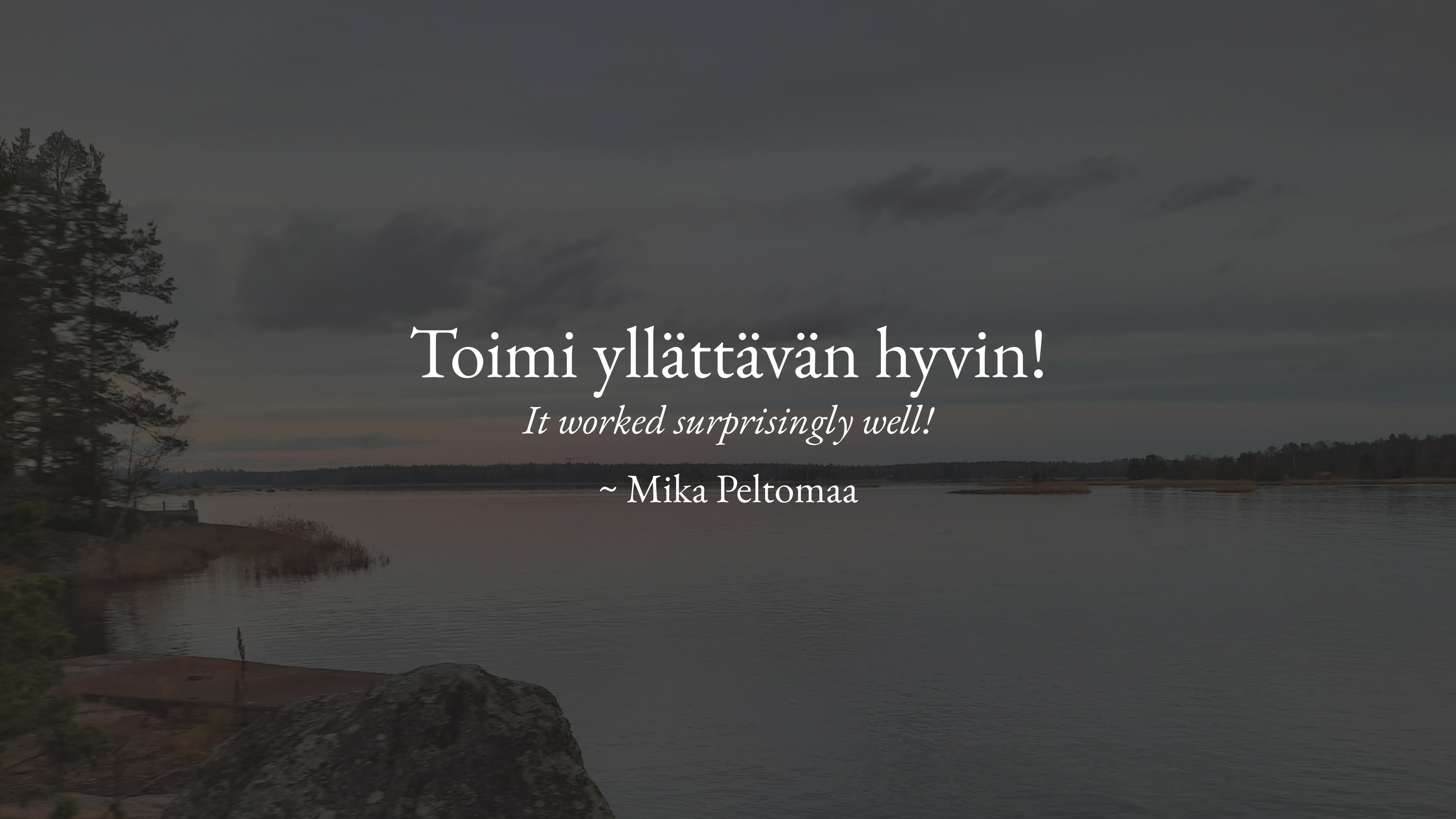
Machine Eval.	BLEU	chrF
Masiingäänttär	**15.14**	**48.87**
GPT-4	9.98	43.66
GPT-3.5	9.11	42.50

BLEU and chrF:
323 segments

MTEval:
50 segments

Human Eval.	Eval 1	Eval 2	Average
Masiingäänttär	1.84	**1.5**	**1.67**
GPT-4	**1.74**	1.94	1.84
GPT-3.5	2.42	2.46	2.44

Anna and Mika Peltomaa
Rauma experts and MTEval'ers


Toimi yllättävän hyvin!
It worked surprisingly well!
~ Mika Peltomaa

Expectations v. Results

- I expected GPT to be able to discern Rauma's morphological differences.
  - **True.** GPT generated accurate translations of morphology, which my model was able to learn.
- I expected using a dictionary to drastically improve GPT's output.
  - **True.** GPT was able to create more accurate and fluent translations when provided with terminology.
- I expected my neural model to outperform GPT.
  - **True.** GPT still does not know enough about Rauma to provide adequate translations.

Future Work

- Data Synthesis
  - Train a Rauma language model
  - More back translation
  - Keep searching for data
  - Rule-based synthesis
  - Dictionary replacements
  - Incorporate other dialectal data
- Translation Models
  - Rauma to Finnish
  - Bidirectional model
  - Finetune OPUS' multilingual models that include Finnish

AI-generated image of Finnish lake with sauna

Questions?

Kysymyksi?